Shorten Up or Swing Away? A Machine-Learning Approach to Two-Strike Hitting

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1. Introduction

Baseball occupies a unique position among American professional sports: it is a game deeply rooted in tradition, one in which conventional wisdom has been passed down across generations with little scrutiny. For much of the sport's history, strategic decision-making at both the team and player level was driven by intuition and experience. However, over the past two decades, the integration of technology and analytics has fundamentally altered much of how the game is played and understood. Advanced statistical modeling and increasingly granular player-tracking data have enabled researchers to reconsider assumptions that were previously accepted as fact – and in many cases, to completely overturn them. This analysis will aim to continue that work, in this case, focused on a batter's approach in 2-strike counts.

A long-standing approach derived from traditional baseball strategy suggests that batters should fundamentally alter their swing mechanics when facing a two-strike count. This conventional wisdom often manifests as instructions to "protect the plate", "shorten up" the swing path, or simply put, just make contact. No matter the verbiage, the idea remains the same – it's better to put the ball in play in a two-strike count. The underlying logic stems from risk aversion: by minimizing the probability of a strikeout, the batter theoretically increases the likelihood of a productive outcome, such as a defensive error or bloop hit.

However, this approach relies on an unverified trade-off between contact frequency and contact quality. If intentionally adjusting a swing to prioritize contact of any kind significantly degrades the batter's exit velocity (how hard the ball was hit) and launch angle (how high the ball was hit), the resulting 'ball in play' may provide less expected value than a standard, high-intent swing – even when accounting for increased strikeout risk.

The traditionalist two-strike approach does appear to dominate Major League Baseball: data from 2023-2025 reveals that batters swung at an average of 72.2 mph in non-two-strike counts versus 70.7 mph in two-strike counts [1]. This 1.5 mph reduction in swing speed suggests a league-wide tendency to sacrifice swing speed in two-strike situations in exchange for an attempted increase in contact likelihood.

By using newly published bat-tracking data from Major League Baseball's Statcast system, this study aims to evaluate the traditionalist two-strike approach, hypothesizing that hitters who maintain a more aggressive swing profile in two-strike counts may generate comparable or superior contact quality to those who shorten up.

2. Data Overview

This analysis utilizes pitch-level data from MLB's Statcast tracking system [1]. Comprised of high-speed cameras and radar systems installed in all thirty Major League ballparks, Statcast has been one of the main drivers of professional baseball's analytical revolution. The technology is designed to capture granular physical movements of in-game events, including pitch movement, swing angle, baserunner speed, among others. MLB publicizes much of this data via their Baseball Savant platform, which serves as the data source for this analysis.

In 2024, Statcast debuted a series of innovative bat tracking metrics [2], widely regarded as a significant advancement by industry experts. These bat tracking metrics are the main driver in understanding how a swing might change with two strikes. Table 1 defines the relevant features needed for this analysis:

Feature	Category	Description
bat_speed	Bat Tracking	Bat head speed through the swing zone (mph)
swing_length	Bat Tracking	Total distance traveled by the bat head in a full swing (ft)
attack_angle	Bat Tracking	Vertical angle of bat barrel at contact (°)
attack_direction	Bat Tracking	Horizontal angle of bat barrel at contact (°); negative = pull side, positive = opposite field
swing_path_tilt	Bat Tracking	Vertical tilt of the full swing arc (°)
launch_speed	Contact Quality	Exit velocity - speed of ball off the bat (mph)
launch_angle	Contact Quality	Vertical angle of batted ball at contact (°)
xwOBA (target variable)	Contact Quality	Expected Weighted On-Base Average; model-estimated hit value based on exit velocity and launch angle, isolating contact quality from defensive outcome
release_speed	Pitch Attributes	Speed of pitch at pitcher's release point (mph)
pfx_x / pfx_z	Pitch Attributes	Horizontal and vertical pitch movement (in)
plate_x / plate_z	Pitch Attributes	Horizontal and vertical coordinates of where the pitch crosses home plate (ft, from center)
balls / strikes	Contextual	Comprises the count; used to identify two-strike situations
description	Contextual	Individual pitch outcome (swinging_strike, foul, hit_into_play, etc.)
events	Contextual	Plate appearance outcome (single, strikeout, home_run, etc.)
stand	Contextual	Batter handedness (R/L)
p_throws	Contextual	Pitcher handedness (R/L)

Table 1 - Competitive Swings Dataset Features sourced via MLB Statcast

The target dataset is comprised of MLB regular season data from the 2024 and 2025 seasons. Pitch-level Statcast data was sourced via a pre-compiled Kaggle dataset [3], which itself was assembled using pybaseball – an open-source Python library providing a programmatic wrapper for Baseball Savant's Statcast API [4]. While this data is publicly available directly through Baseball Savant, the pre-compiled Kaggle dataset was used for convenience. Supplementary player information, including batter names, was retrieved directly via pybaseball's *playerid_reverse_lookup* function to map Statcast's numeric batter ID to readable names [4].

2.1. Data Preprocessing

To prepare for analysis and modeling, the raw dataset was cleaned and filtered extensively. Initially, any non-swing events were removed from the dataset, retaining only pitches on which the batter attempted a swing. Bunt attempts were subsequently removed, as this analysis requires full, high-intent swings – bunt mechanics are not representative of a batter's natural swing profile.

Building on this, a competitive swing filter was applied using Statcast's published definition: the fastest 90% of each player's swings, as well as any 60+ mph swings resulting in an exit velocity of

90+ mph [5]. This filter removes check-swings and defensive swings that would otherwise introduce noise into swing profile measurements. The competitive swing filter is preferable to a blanket speed floor that would not account for natural variation in swing speed across player profiles. In total, the competitive swing filter removed 60,609 swings (9.6% of the dataset).

Additionally, all switch hitters were removed from the dataset. Switch hitters present an interpretability challenge for the attack direction feature, as the sign convention (negative indicating pull side, positive indicating opposite field) reverses depending on which side of the plate the batter is standing, posing data accuracy issues. In total, 67 switch hitters accounting for approximately 72,000 pitches were removed, with notable exclusions including José Ramírez, Francisco Lindor, and Cal Raleigh. Accommodating switch hitters through a handedness-stratified approach is identified as an area for future work.

Lastly, a minimum sample threshold of 40 two-strike competitive swings per batter was enforced, ensuring that each batter's delta features reflect a sufficient number of observations to capture true tendencies rather than small-sample noise. Any remaining contact events with missing launch speed or launch angle values, likely due to Statcast failures, were dropped, removing 548 events (0.1% of the dataset). The final dataset was comprised of 505,843 competitive swings across 551 batters.

3. Swing Profile Identification via Unsupervised Clustering (Phase 1)

3.1. Feature Engineering

Before quantifying the outcome of each two-strike approach, it's imperative to actually define the approaches themselves based on swing adjustments captured by the bat tracking data. To engineer features that quantify these swing adjustments, the competitive swings dataset was partitioned into two-strike ($strikes\ =\ 2$) and non-two-strike ($strikes\ <\ 2$) subsets. A batter-specific mean was computed for each of the five swing tracking features in both subsets, and the delta between the two was calculated as follows:

$$\Delta_{feature} = \bar{x}_{feature,strikes=2} - \bar{x}_{feature,strikes<2}$$

This produced five delta features per batter - $\Delta_{bat\ speed}$, $\Delta_{swing\ length}$, $\Delta_{attack\ angle}$, $\Delta_{attack\ direction}$, and $\Delta_{swing\ path\ tilt}$ – each representing the magnitude and direction of a batter's two-strike adjustment relative to their baseline swing profile. A negative delta indicates a reduction in that metric in two-strike counts, while a positive delta indicates an increase. For example, during his 2025 rookie season, Atlanta Braves catcher Drake Baldwin recorded a $\Delta_{swing\ length}$ of +0.22 feet. This indicates that, contrary to the traditional approach of "shortening up" with two strikes, Baldwin's swing length actually increased on average.

Using deltas instead of raw averages is critical here as it shifts the focus from a player's baseline to their actual tendency to adjust with two strikes. This analysis cares less about who swings the fastest, and more about who chooses to swing faster (or slower) when they're down in the count. The goal is to make sure clustering is done based on adjustment behavior, rather than just grouping batters by their natural swing profiles.

The resulting batter-level dataset consists of one observation per player, allowing for analysis of 551 swing profiles.

3.2. Principal Component Analysis – The Dimensions of Adjustment

Using K-Means Clustering, batters are to be categorized according to these adjustments, aiming to reveal distinct characterizations of how different hitters adjust when facing a strikeout threat.

However, prior to clustering, exploratory analysis revealed strong multicollinearity among several features, as shown in Figure 1. This result was to be expected, given each feature quantifies one dimension of an interconnected swing. As an example, a batter who shortens their swing ($\Delta_{swing\ length}$) will also tend to flatten their attack angle ($\Delta_{attack\ angle}$), a relationship reflected through a correlation of $r = 0.75$ between the two metrics.

Figure 1 - Correlation Matrix of Delta Swing Features

To address this redundancy, Principal Component Analysis (PCA) was applied. By transforming the five delta features into a smaller set of orthogonal components, variance is still preserved while eliminating redundant dimensions.

PCA was fit on standardized delta features, with standardization applied prior to decomposition to ensure no single feature dominated the components due to differences in scale. The results yielded three components with clear, interpretable meanings in a baseball context. Together, these components account for 92.2% of the total variance in two-strike adjustments. The explained variance by component is shown in Table 2, and the component loadings are presented in Table 3.

Principal Component 1: Swing Aggression (57.0% of variance)

PC1 is characterized by strong loadings for $\Delta_{swing\ length}$ $(+0.536)$, $\Delta_{attack\ angle}$ $(+0.545)$, $\Delta_{attack\ direction}$ (-0.509), and $\Delta_{bat\ speed}$ $(+0.390)$. PC1 scores can be interpreted in one of two ways:

- **Positive PC1 Score:** Profiles a "power-seeking" adjustment. These hitters increase their swing length and bat speed while aiming to pull the ball in the air.

- **Negative PC1 Score:** Profiles a "contact-oriented" adjustment. These hitters prioritize putting the ball in play by shortening the swing, flattening the plane, and hitting to the opposite field.

Principal Component 2: Independent Plane Adjustment (21.3% of variance)

PC2 is heavily dominated by $\Delta_{swing\ path\ tilt}$ $(+0.907)$. This confirms the independence observed in the initial correlation analysis; adjustments to the vertical orientation (tilt) of the swing path appear to be a distinct mechanical choice, largely separated from the length or speed of the swing itself.

Principal Component 3: Residual Speed (13.9% of variance)

PC3 is primarily defined by $\Delta_{bat\ speed}$ $(+0.786)$. This component suggests that a significant portion of bat speed variation happens independently of the swing compression captured in PC1. Hitters may alter their effort level (speed) without actually changing the geometry (length or angle) of their swing.

Component	Explained Variance	Cumulative Variance
PC1	57.0%	57.0%
PC2	21.3%	78.3%
PC3	13.9%	92.2%
PC4	4.4%	96.7%
PC5	3.3%	100.0%

Table 2 - Explained variance by principal component. Three components were retained for clustering, accounting for 92.2% of total variance.

Feature	PC1	PC2	PC3
Δ Bat Speed	0.390	0.327	0.786
Δ Swing Length	0.536	0.159	-0.016
Δ Attack Angle	0.545	-0.111	-0.179
Δ Attack Direction	-0.509	0.181	0.444
Δ Swing Path Tilt	-0.066	0.907	-0.391

Table 3 - Component loadings for retained components. Loadings represent the correlation between the original delta feature and the corresponding principal component.

3.3. K-Means Clustering

With the three principal components serving as input features, K-means clustering was employed to identify distinct batter archetypes. The optimal number of clusters (k) was selected by evaluating inertia and silhouette scores across k values from k=2 through k=10, shown in Figure 2.

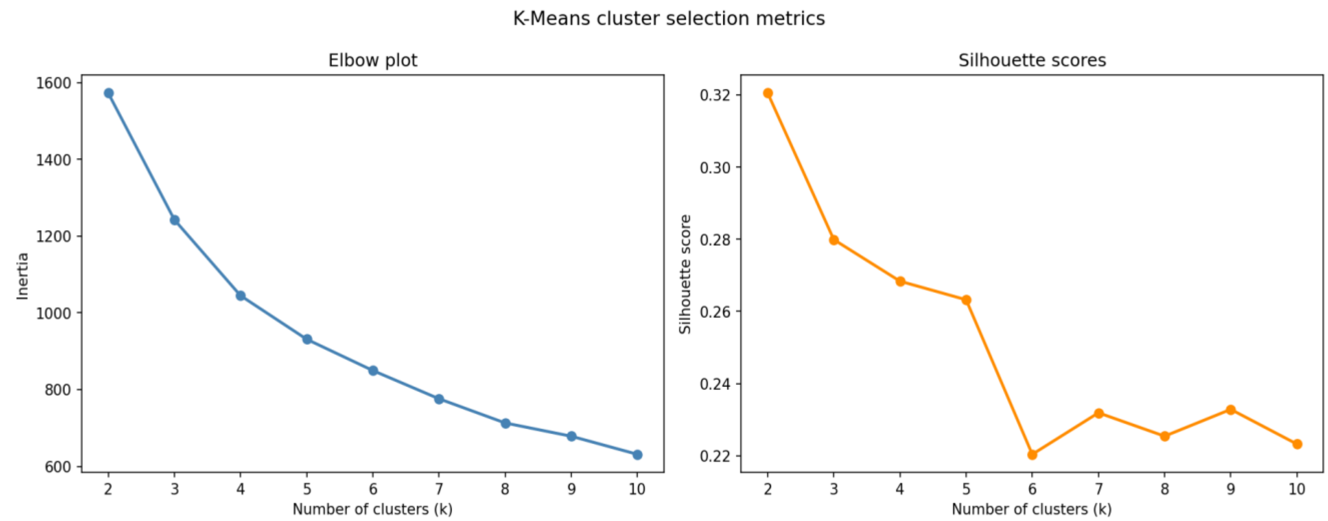

Figure 2 - Elbow Plot and Silhouette Scores used to identify optimal k for clustering

The elbow plot identified k=3 as the point of diminishing returns, where the rate of inertia reduction slowed substantially. While the silhouette score was found to be highest at k=2 (0.32), a binary classification proved insufficient for the study's objectives. A two-cluster solution essentially reduces the population to 'adjusts' vs. 'does not adjust', failing to capture the nuanced mechanics of a swing profile.

Because k = 3 yielded the next highest silhouette score (0.28) and aligned with the elbow plot results, it was selected as the final solution, providing an ideal balance between statistical robustness and interpretable baseball insight. The three resulting archetypes – Free Swingers, Moderate Adjustors, and Traditional Shorteners – were named based primarily on the ordering of the $\Delta_{swing\ length}$ centroids. Full centroid profiles for each archetype are presented in Table 4 and visualized in Figure 3.

Archetype	**Δ Bat Speed (mph)**	**Δ Swing Length (ft)**	**Δ Attack Angle (°)**	**Δ Attack Direction (°)**	**Δ Swing Path Tilt (°)**
Free Swinger (n=160)	-0.92	0.037	0.72	-1.623	-0.771
Moderate Adjustor (n=282)	-1.387	-0.093	-0.968	1.118	-0.626
Traditional Shortener (n=109)	-2.178	-0.241	-2.704	3.651	-0.539

Table 4 - Cluster centroids expressed in original feature space. Negative values indicate a reduction in two-strike counts; positive values indicate an increase.

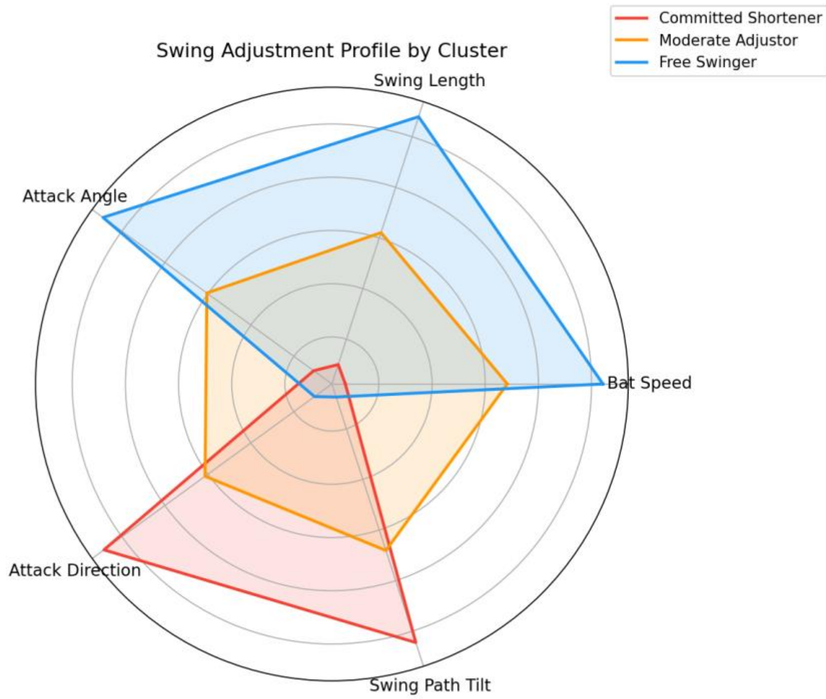

Figure 3 - Radar Chart visualizing swing adjustment profiles by archetype

a. Free Swingers (n=160)

The Free Swingers are represented by the smallest two-strike adjustments across all features, resisting the traditionalist approach of shortening up. Notably, they show marginally increasing swing length (+0.037 ft) and attack angle (+0.72°), while recording the smallest bat speed reduction (-0.92 mph). Additionally, this is the only profile with a negative attack direction (-1.623°), indicating a tendency to pull the ball. This archetype profiles a batter looking to hit for power regardless of the count – maintaining an aggressive swing to the pull side while attempting to elevate the baseball.

Domain validation of this cohort includes several high-pedigree rookies from the 2025 season, such as Roman Anthony, Marcelo Mayer, and Drake Baldwin. The presence of these prospects suggests a developmental trend: younger hitters may rely more heavily on their "baseline" mechanics in high-leverage counts, potentially due to a lack of professional experience in executing situational adjustments. This cluster also features established veterans known for high-variance, power-focused profiles, most notably Giancarlo Stanton and Joey Gallo, who traditionally maintain high intent levels even at the risk of increased strikeout rates.

b. Traditional Shorteners (n=109)

Traditional Shorteners (n=109) represent the opposite extreme, embodying the traditional "protect the plate" philosophy. This group shows the most dramatic reductions in swing length (-0.241 ft), bat speed (-2.18 mph), and attack angle (-2.70°). The significantly positive attack direction delta (+3.651°) shows a deliberate approach to hit the ball to the opposite field. For these hitters, the two-strike adjustment is a fundamental mechanical overhaul aimed at maximizing the probability of contact. Domain validation confirms this archetype; the cluster is populated by technical hitters like Steven Kwan and J.D. Martinez, who are recognized league-wide for their ability to prioritize contact and situational hitting over raw power.

c. Moderate Adjustors (n=282)

The largest group in the analysis, Moderate Adjustors (n=282) occupies the middle ground of the approach spectrum. While they do not completely shift their mechanics like the Shorteners, they make meaningful, controlled concessions in exchange for contact. Their reductions in bat speed (-1.39 mph) and swing length (-0.09 ft) are consistent but less severe, representing a balanced approach that attempts to mitigate the strikeout threat without completely sacrificing their baseline power profile.

Most significantly, some of the league's most elite offensive performers, Aaron Judge, Shohei Ohtani, Juan Soto, and Yordan Álvarez, clustered as Moderate Adjustors. This suggests two possible conclusions:

a. The most effective two-strike approach is a measured, controlled adjustment that mitigates the strikeout threat without compromising the core mechanical efficiency that makes these hitters elite.

b. This tier of elite hitters possess such high baseline contact ability that they are not required to "over-adjust", which might otherwise reduce their overall production.

The fact that the game's premier hitters utilize a moderate rather than extreme adjustment provides a strong foundation for the subsequent analysis of outcome metrics.

4. Outcome Prediction via Supervised Learning (Phase 2)

With the batter archetypes established, Phase 2 aims to quantify whether meaningful differences in contact quality exist across the three different approaches. To do so, a series of regression models were trained to predict Expected Weighted On-Base Average (xwOBA) on two-strike contact events.

Developed by MLB Statcast, xwOBA is an evolution of Weighted On-Base Average (wOBA), a metric that assigns relative values to each hit type based on its actual run-scoring potential (e.g., a double is worth more than a single, but not twice as much) [6]. While wOBA measures realized results, xwOBA isolates the batter's performance from external factors like defensive positioning and ballpark factors. It is calculated by mapping the Exit Velocity and Launch Angle of a batted ball against historical outcomes for balls with similar physical characteristics.

The use of xwOBA is critical for this analysis for two reasons:

- **Isolation of Contact Quality:** A batter who hits a 105 mph line drive directly at an infielder is "punished" by box-score metrics with an out (wOBA = 0). However, this is considered nothing more than bad luck, as the swing itself was elite. xwOBA assigns that event a high value (e.g., 0.850) based on the likelihood of it being a hit, rewarding the quality of the swing rather than the luck of the result.
- **Scale and Interpretability:** Each ball in play generates a discrete xwOBA value on a scale that mirrors On-Base Percentage (OBP). In the modern MLB landscape, a season long

average xwOBA typically sits around .320, while elite contact quality is represented by values exceeding .400 [7]. This allows for a granular, pitch-level assessment of how different mechanical archetypes influence the expected value of a contact event.

4.1. Dataset and Feature Selection

The Phase 2 dataset was constructed by filtering the competitive swings dataset to include only two-strike contact events. Specifically, this is pitches where the batter made contact and a valid xwOBA value was recorded. This restriction is necessary because xwOBA is defined exclusively for balls in play; swinging strikes and foul balls do not produce a contact quality measurement.

The final regression dataset consisted of 71,079 pitch-level contact events across the same 551 unique batters. The archetype labels from Phase 1 were merged with this dataset, resulting in the following distribution of contact events:

- **Free Swingers:** $n = 18{,}879$
- **Moderate Adjustors:** $n = 37{,}591$
- **Traditional Shorteners:** $n = 14{,}609$

Two categories of variables were used as model inputs to ensure the predicted outcomes isolated the effect of the batter's mechanics:

1. **Bat Tracking:** The five core tracking features – Bat Speed, Swing Length, Attack Angle, Attack Direction, and Swing Path Tilt – were used to capture the batter's physical mechanics on each specific contact event.
2. **Pitch Attributes:** Release Speed, horizontal movement (pfx_x), vertical movement (pfx_z), and pitch location (plate_x, plate_z) were included as control variables. This accounts for the variance in contact quality attributable to the quality of pitch thrown, independent of the batter's swing behavior.

4.2. Model Training

Three regression models were trained and evaluated: a baseline Linear Regression, Random Forest $(n\_estimators = 100, max\_depth = 6)$, and XGBoost $(n\_estimators = 100, max\_depth = 6, learning\_rate = 0.1)$. The tree-based hyperparameters were selected to balance model complexity with computational efficiency rather than through exhaustive tuning, which is acknowledged as a limitation. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used as evaluation metrics, both of which maintain interpretability on the xwOBA scale.

A critical choice in the cross-validation setup was the use of 5-fold GroupKFold cross-validation, using batter as the grouping variable. This player-split approach ensures that no batter appears in both the training and test sets within any given fold, preventing the model from memorizing individual player tendencies. A standard random split at the pitch level would violate this independence assumption, as the same batter's swings would appear in both training and test sets across folds. Cross-validation results are presented in Table 5.

Model	RMSE Mean	RMSE Std	MAE Mean	MAE Std
Linear Regression	0.3556	0.0049	0.2778	0.0035
Random Forest	0.3522	0.0048	0.2747	0.0031
XGBoost	0.3512	0.0049	0.2725	0.0034

Table 5 - Five-fold GroupKFold cross-validation results for all three regression models.

All three models produced comparable RMSE values, ranging from 0.351 to 0.356, suggesting the relationship between swing profile features and contact quality is largely captured by linear associations. XGBoost was selected as the final model based on its marginally superior RMSE (0.3512) and MAE (0.2725) across all five folds, as well as its ability to provide feature importance rankings that offer better insight into the relative contribution of each input feature.

5. Results – The Impact of Approach on Performance

With the archetypes defined and the predictive model validated, this final section evaluates the relationship between a batter's two-strike approach and their resulting contact quality.

5.1. Feature Importance on Contact Quality

The XGBoost feature importance rankings (Figure 4) identify Bat Speed as the primary driver of contact quality in two-strike counts, accounting for 22.9% of total importance. Pitch location metrics plate_x (13.3%) and plate_z (8.8%) collectively follow at 22.1%, validating the inclusion of pitch-level covariates.

Notably, directional features like Attack Direction (10.7%) and Attack Angle (10.6%) significantly outperform Swing Length (6.2%), which ranked eighth of ten features. This finding presents a direct contradiction to traditional baseball wisdom: while "shortening the swing" is the most common instruction in two-strike counts, it contributes the least to predicted contact quality. Conversely, maintaining bat speed is the strongest individual predictor of success.

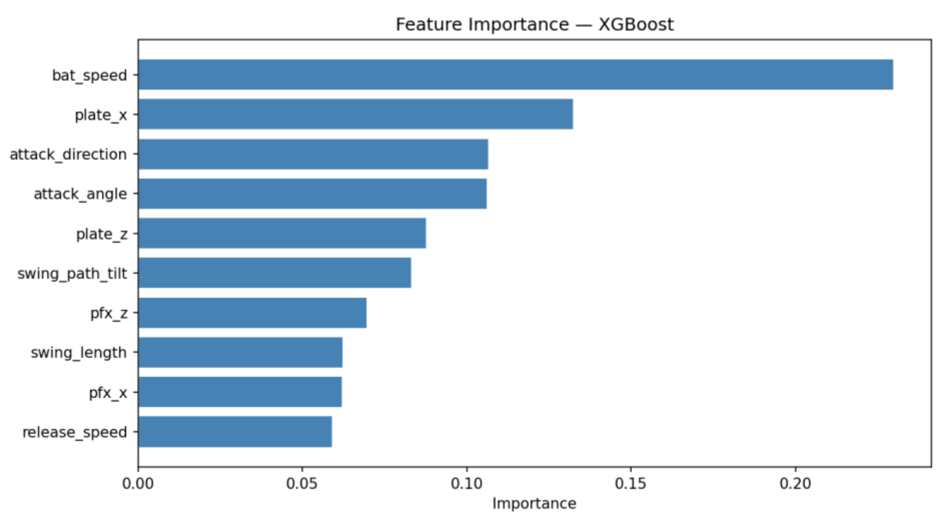

Figure 4 - Feature Importance Chart determined by finalized XGBoost Regressor Model

5.2. Comparative Analysis of Archetypes

To determine if these mechanical adjustments translate to statistically significant outcomes, mean predicted xWOBA was compared across the three archetypes. Results are summarized in Table 6.

Archetype	Mean xwOBA	Std. Dev	n (Batters)
Free Swinger	0.3655	0.0365	160
Moderate Adjustor	0.3566	0.038	282
Traditional Shortener	0.3458	0.0367	109

Table6 - Mean Predicted xwOBA by Archetype

A Kruskal-Wallis test confirmed a highly significant difference in predicted xwOBA between the groups ($H = 19.22, p < 0.001$). Post-hoc analysis using the Games-Howell test (Table 7) revealed that all pairwise comparisons were statistically significant at $\alpha = .05$, establishing a clear hierarchy of contact quality:

1. **Free Swinger vs. Traditional Shortener** ($\boldsymbol{p < 0.001}$): The largest performance gap observed. Free Swingers produced an average xwOBA nearly 0.0197 points higher than those who fully committed to shortening their swing. This difference is in fact meaningful in a baseball context, where single-digit wOBA point differences between hitter tiers result in wide gaps in run production across a full season.
2. **Moderate Adjustor vs. Traditional Shortener** ($\boldsymbol{p = 0.029}$): Even a measured adjustment outperformed the more extreme defensive approach.
3. **Free Swinger vs. Moderate Adjustor** ($\boldsymbol{p = 0.041}$): Despite the Moderate cluster containing many of the league's most elite hitters, the Free Swinger approach maintained a statistically superior profile for pure contact quality.

Comparison	Mean Difference (xwOBA)	p-value
Free Swinger vs. Traditional Shortener	0.0197	<0.001
Moderate Adjustor vs. Traditional Shortener	0.0108	0.029
Free Swinger vs. Moderate Adjustor	0.0089	0.041

Table 7 - Games-Howell post-hoc pairwise comparisons of mean predicted xwOBA across archetypes. Positive mean differences indicate the first-named archetype produced higher predicted xwOBA.

The results suggest a significant performance tax associated with traditional two-strike adjustments. While Traditional Shorteners prioritize the probability of contact (minimizing strikeouts), the mechanical concessions required result in the lowest expected contact quality of any group.

In contrast, Free Swingers maintain their baseline mechanics, which yields the highest predicted xwOBA. This suggests that for many hitters, the most effective "adjustment" may be no adjustment at all. The Moderate Adjustors appear to occupy a functional middle ground, perhaps sacrificing a marginal amount of contact quality (0.009 xwOBA vs. Free Swingers) to achieve the elite overall production seen in players like Shohei Ohtani and Aaron Judge.

6. Conclusion

This study utilized a two-phase analytical approach to quantify the mechanical nature and performance impact of two-strike hitting adjustments in professional baseball. By isolating batter-specific swing "deltas," three distinct archetypes were identified: Free Swingers, Moderate Adjustors, and Traditional Shorteners.

One of the more significant findings of this analysis is the disconnect between traditional coaching wisdom and empirical contact quality. For decades, hitters have been instructed to "shorten up" with two strikes to prioritize contact. However, our Phase 2 modeling revealed that Swing Length is the least influential predictor of expected contact quality xwOBA, while Bat Speed remains the most critical driver of success.

The data suggests that the "Traditional Shortener" approach imposes a significant performance tax. These hitters produced the lowest predicted xwOBA (0.3458), trailing the aggressive "Free Swinger" profile by nearly 20 points. While shortening the swing may increase the probability of contact (reducing strikeouts), it fundamentally degrades the quality of that contact when it occurs.

The distribution of the league's most elite performers, such as Shohei Ohtani and Aaron Judge, into the Moderate Adjustor cluster suggests an elite and optimal approach to two-strike hitting. These players do not maintain the aggression of Free Swingers, nor do they sell out, making the extreme mechanical sacrifices of the Shorteners. Instead, they utilize a calculated adjustment that preserves the majority of their bat speed and swing path while mitigating the risks inherent in two-strike counts.

Ultimately, this analysis demonstrates that in the modern era of high-velocity pitching and optimized defensive positioning, "protecting the plate" should not be synonymous with "abandoning the swing." Maintaining a high-intent, speed-driven profile, even with two strikes, is statistically superior to the traditional defensive approach. As player development continues to lean into data-driven strategies, these findings suggest that the next generation of hitters may find more success by trusting their baseline mechanics rather than fundamentally altering them when the count is against them.

7. References

[1] MLB Baseball Savant. *Statcast Search*.
https://baseballsavant.mlb.com/statcast_search/

[2] MLB.com. What you need to know about Statcast bat tracking.
https://www.mlb.com/news/what-you-need-to-know-about-statcast-bat-tracking

[3] Yasunorim. *MLB Statcast Bat Tracking 2024–2025* [Data set]. Kaggle.
https://www.kaggle.com/datasets/yasunorim/mlb-statcast-bat-tracking-2024-2025

[4] Langtry, J. *pybaseball*. [Software]. GitHub. https://github.com/jldbc/pybaseball

[5] MLB Baseball Savant. Bat Tracking Leaderboard.
https://baseballsavant.mlb.com/leaderboard/bat-tracking

[6] FanGraphs Sabermetrics Library. Weighted On-Base Average (wOBA).
https://library.fangraphs.com/offense/woba/

[7] MLB Baseball Savant. Expected Statistics (xwOBA) Leaderboard.
https://baseballsavant.mlb.com/leaderboard/expected_statistics